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Uninterruptible power supply having a liquid cooling device

Abstract

A liquid cooling device for cooling at least one target component that includes: a base member to be in thermal contact with the at least one target component to be cooled by the liquid cooling device, the base member including a lower surface configured to be placed in contact with the at least one target component, the base member defining a plurality of pockets spaced apart from one another on an upper side thereof and defining a plurality of fluid conduits configured to internally circulate a cooling liquid therethrough. Each of the plurality of pockets arranged to accommodate a corresponding fluid conduit and an interconnecting channel defined by the base member and extending between two neighboring pockets of the plurality of pockets for distributing the cooling fluid from the fluid conduit of one pocket to the fluid conduit of a neighboring pocket.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a Continuation-In-Part (CIP) application of U.S. patent application Ser. No. 17/308,181 filed on May 5, 2021 which, in turn, claims priority from European Pat. Appln. No. 20315286.3 filed on May 29, 2020. The present application also claims priority from European Pat. Appln. 22306286.0, filed Aug. 30, 2022. The entirety of each of the above-referenced documents is incorporated herein by reference.

FIELD OF TECHNOLOGY
(1) The present technology relates to uninterruptible power supplies and, in particular, to those used in data centers.
BACKGROUND
(2) Uninterruptible power supplies (UPSs) are used to provide backup power to a target load in case a main electrical supply thereof should fail or otherwise be subject to a change that could negatively affect the target load. In particular, if the main electrical supply to the target load is interrupted or otherwise affected, a UPS connected to the target load would immediately take over supplying electricity without any interruption to the system. For that reason, UPSs are an important component in data centers, namely to ensure continuous operation of servers and other equipment housed in a data center (e.g., cooling equipment).
(3) While UPSs are undoubtedly useful, they also typically generate significant amounts of heat which must be taken into consideration in their design and in their environmental surroundings (e.g., their placement within a data center). Many heat management solutions exist for UPSs, including for instance providing dedicated air cooling equipment (e.g., an air handling unit) between two UPSs to allow them to aspirate air discharged by the air cooling equipment, or for instance providing separator walls for each UPS to ensure that cool air is aspirated into the UPS. However, these solutions can require significant resources to implement, and moreover can occupy

considerable space within the data center. For instance, in some cases, raised floors and/or false ceilings may be implemented to provide a route for heated or cooled air to circulate therethrough.
(4) Therefore, there exists an interest in a UPS configuration that can ameliorate at least some of the noted drawbacks.

SUMMARY

(5) It is an object of the present technology to ameliorate at least some of the drawbacks of the prior art UPS configurations.
(6) According to one aspect of the present technology, there is provided a liquid cooling device for cooling at least one target component including a base member in thermal contact with the at least one semiconductor target component, the base member including a lower surface placed in contact with the at least one semiconductor target component and an upper surface containing a plurality of pockets spaced apart from one another. The plurality of pockets arranged to accommodate a fluid conduit structure configured to internally circulate a cooling liquid therethrough; and an interconnecting channels formed between neighboring pockets of the plurality of pockets and configured to distribute the cooling liquid from the fluid conduit structure of one pocket to the fluid conduit structure of a neighboring pocket. The fluid conduit structures interconnected via the internal interconnecting channels are alignedly positioned with the at least one semiconductor for cooling thereof.
(7) In some embodiments, the fluid conduit structures comprise a fluid inlet to receive the cooling liquid and a fluid outlet to discharge the circulated cooling liquid, in which the fluid outlet of a fluid conduit structure of a first neighboring pocket distributes the circulated liquid to the fluid inlet of a fluid conduit structure of a second neighboring pocket via the internal interconnecting channel.
(8) In some embodiments, the interconnecting channel is formed between adjacent sidewalls of respective neighboring pockets.
(9) In some embodiments, the interconnecting channel comprises a slanted-shaped profile in which one end of the interconnecting channel is disposed vertically higher than the other end of the interconnecting channel.
(10) In some embodiments, the interconnecting channel comprises a V-shaped profile in which both ends of the interconnecting channel are disposed vertically higher than a downwardly-angled vertex point.
(11) In some embodiments, the plurality of pockets comprises an even number of pockets arranged along a two dimensional plane of the upper surface of the base member.
(12) In some embodiments, the pockets incorporate the internal interconnecting channel to service neighboring pockets disposed along a first dimension of the upper surface and incorporates the interconnecting channel to service neighboring pockets disposed along a second dimension of the upper surface.
(13) In some embodiments, the upper surface of the base member comprises an overhead fluid inlet tube, disposed along a top surface of the upper surface, for supplying the cooling liquid to the pockets from a liquid source and an overhead fluid outlet tube, disposed along a top surface of the upper surface, for expelling the circulated liquid back to the liquid source.
(14) In some embodiments, the upper surface of the base member comprises a lateral fluid inlet tube, disposed along a side surface of the upper surface, for supplying the cooling liquid to the pockets from a liquid source and an overhead fluid outlet tube, disposed along a side surface of the upper surface, for expelling the circulated liquid back to the liquid source.
(15) In some embodiments, the interconnecting channel is formed by a milling or drilling process.
(16) Embodiments of the present technology each have at least one of the above-mentioned objects and/or aspects, but do not necessarily have all of them. It should be understood that some aspects of the present technology that have resulted from attempting to attain the above-mentioned object may not satisfy this object and/or may satisfy other objects not specifically recited herein.
(17) Additional and/or alternative features, aspects and advantages of embodiments of the present

technology will become apparent from the following description, the accompanying drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of the present technology, as well as other aspects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:
(2) FIG. **1** is a perspective view, taken from a top, right side, of an uninterruptible power supply (UPS) according to an embodiment of the present technology;
(3) FIG. **2** is a perspective view, taken from a top, left side, of the UPS of FIG. **1**;
(4) FIG. **3** is a front elevation view of the UPS of FIG. **1**;
(5) FIG. **4** is a right side elevation view of the UPS of FIG. **1**;
(6) FIG. **5** is a top plan view of the UPS of FIG. **1**;
(7) FIG. **6** is a schematic illustration of an electrical system of the UPS of FIG. **1**;
(8) FIG. **7** is a perspective view, taken from a top, left side, of the UPS of FIG. **1**, shown with various walls of a housing of the UPS removed and with a door of the UPS being open;
(9) FIG. **8** is a perspective view, taken from a top, right side, of the UPS of FIG. **7**;
(10) FIG. **9** is a perspective view, taken from a top, left side, of part of the UPS of FIG. **7**, showing a plurality of liquid cooling devices of the UPS;
(11) FIG. **10** is a top plan view of the part of the UPS of FIG. **9**;
(12) FIG. **11** is a perspective view of part of a pumping module of the UPS of FIG. **1**;
(13) FIG. **12** is a schematic representation of a plate heat exchanger of the pumping module of FIG. **11**;
(14) FIG. **13** is a diagram of a liquid cooling circuit and an air cooling circuit of the UPS of FIG. **1**;
(15) FIG. **14** is a front elevation view of an air-to-liquid heat exchanger of the UPS of FIG. **1**,
(16) FIG. **15** is a perspective view of a booster bridge and three rectifier/inverter bridges of the UPS, with a liquid cooling device being mounted to each bridge;
(17) FIG. **16** is a front elevation view of the bridges of FIG. **15**;
(18) FIG. **17** is a perspective view of one of the liquid cooling devices of the UPS of FIG. **1**;
(19) FIG. **18** is a front elevation view of the liquid cooling device of FIG. **17**;
(20) FIG. **19** is a bottom plan view of the liquid cooling device of FIG. **17**;
(21) FIG. **20** is an exploded view of the liquid cooling device of FIG. **17**, shown with three semiconductors to be cooled by the liquid cooling device;
(22) FIG. **21** is a perspective view of a base member of the liquid cooling device of FIG. **17**;
(23) FIG. **22** is a top plan view of the base member of FIG. **21**;
(24) FIG. **23** is a top plan view of the base member of FIG. **21**, shown with the underlying semiconductors in dashed lines;
(25) FIG. **24** is an exploded view of a liquid cooling device according to an alternative embodiment,
(26) FIGS. **25**A, **25**B respectively depict perspective and cross-sectional views of an internal interconnecting channel configuration between liquid cooling elements;
(27) FIGS. **26**A, **26**B respectively depict perspective and cross-sectional views of an internal an alternative interconnecting channel configuration between liquid cooling elements;
(28) FIG. **27** depicts a liquid cooling base member incorporating multiple liquid cooling elements with interconnecting channels; and
(29) FIG. **28** depicts the liquid cooling base member of FIG. **27** incorporating inlet tube and outlet tube connective portions disposed on a lateral side of the base member.

DETAILED DESCRIPTION

(30) FIGS. **1** to **5** show an uninterruptible power supply (UPS) **10** in accordance with an embodiment of the present technology. In use, the UPS **10** is electrically connected to a target load **13** (see FIG. **6**) so as to provide backup electrical power thereto to ensure continuous operation of the target load **13** in case a primary power source **17** (e.g., an electrical network) that provides electrical power for powering the target load **13** fails or otherwise is subject to discontinuous performance (e.g., spikes in voltage). The UPS **10** can be used in different types of applications and therefore the target load **13** can constitute different types of equipment depending on the application. For instance, in this particular example, the UPS **10** is used in a data center **100** for providing backup power to data center equipment housed within the data center **100**. Thus, in this example, the target load **13** is data center equipment which can include but is not limited to servers, network equipment, cooling equipment, etc.

(31) As will be described in greater detail below, the UPS **10** is cooled at least in part by liquid cooling devices that are fed and discharged via at least one pumping module of the UPS **10**. This can help in cooling the UPS **10** more efficiently while maintaining safe operating conditions thereof. Moreover, as will be discussed further below, it can also provide more freedom in the configuration of the data center **100** as fewer design restrictions may apply to the UPS **10** than a conventional UPS.

(32) As shown in FIG. **1**, the UPS **10** has a housing **12** defining an interior space **15** within which the various electrical components of the UPS **10** are disposed. The housing **12** has left and right walls **14**, a top wall **16**, a bottom wall **18** and a rear wall (not shown) which together define the interior space **15** of the housing **12**. The top wall **16** and the bottom wall **18** extend generally horizontally while the left and right walls **14** and the rear wall extend generally vertically. The housing **12** also includes a frame **20** (shown in more detail in FIGS. **7** and **8**) for supporting the different walls of the housing **12**, as well as the various electrical components of the UPS **10**. For instance, in this embodiment, the frame **20** includes four vertical beams **22** (one disposed at each corner of the UPS **10**), three upper and three lower lateral beams **24** each extending laterally between two of the vertical beams **22**, and two upper and two lower transversal beams **26** that extend transversally to the vertical and lateral beams **22**, **24**. Each transversal beam **26** is connected between two of the vertical beams **22**. The frame **20** also includes four legs **27**, each connected to a lower end of a corresponding one of the vertical beams **22**. The legs **27** are configured to be supported by a support surface (e.g., the floor). Moreover, the legs **27** elevate the bottom wall **18** off the support surface. The housing **12** and its frame **20** may be configured in any other suitable way in other embodiments.

(33) The interior space **15** of the housing **12** is accessible via two doors **30** disposed on a front side of the UPS **10** and operatively connected to the housing **12**. The doors **30** are selectively opened and closed to access the interior space **15**. In particular, the doors **30** are hinged to the housing **12** such that each door **30** can be pivotably opened and closed about respective hinges **32** defining a hinge axis extending vertically. Each door **30** is generally rectangular and has an interior side **34** and an exterior side **36**. The interior side **34** faces the interior space **15** when the door **30** is closed. One of the doors **30** has a lock **36** which is actuated via a key (not shown) when the doors **30** are closed to lock or unlock the doors **30**. Such locks are known and therefore will not be described in greater detail herein.

(34) The doors **30** support pumping modules **60** and air-to-liquid heat exchangers **90** which assist in cooling the UPS **10**. Notably, the pumping modules **60** and air-to-liquid heat exchangers **90** are mounted to the doors **30** so that the UPS **10** is cooled in a more autonomous manner than in conventional UPSs. The pumping modules **60** and the air-to-liquid heat exchangers **90** will be described further below.

(35) As shown in FIG. **6**, the UPS **10** has an electrical system **40** enclosed within the housing **12**, namely in the interior space **15**. The electrical system **40** includes a plurality of electrical

components which work together to ensure the operation of the UPS **10** to provide backup power to the target load. In this embodiment, the UPS **10** is a double conversion online UPS. Notably, the electrical system **40** includes a rectifier module **42** configured to receive electrical power from the primary power source **17**, an inverter module **44** configured to supply electrical power to the target load **15**, and an internal static bypass switch **46** configured to bypass the rectifier module **42** and the inverter module **44**.

(36) The rectifier module **42** is electrically connected to a plurality of batteries **45** for charging thereof. In this embodiment, the batteries **45** are disposed outside of the UPS **10**, namely in a room adjacent to the room of the data center **100** within which the UPS **10** is located. In other embodiments, the batteries **45** may be part of the electrical system **40** of the UPS **10** and enclosed within the interior space **15** of the housing **12**. The rectifier module **42** charges the batteries **45** by converting the alternating current (AC) electrical charge provided by the primary power supply **17** to direct current (DC) electrical charge that is used as an input to charge the batteries **45**. On the other hand, the inverter module **44** is electrically connected to the batteries **45** for supplying electrical power from the batteries **45** to the target load **13**. In particular, the inverter module **44** is configured to convert the DC electrical charge of the batteries **45** to AC electrical charge to supply power to the target load **13**. For its part, the internal static bypass switch **46** allows instantaneously bypassing the rectifier module **42**, the inverter module **44** and the batteries **45**, so as to power the target load **13** directly via the primary power source **17**, for instance when there is an internal fault or failure within the electrical system **40** of the UPS **10**. This can thus ensure power continuity to the target load **13** while the UPS **10** is being fixed.

(37) As shown in FIGS. **15** and **16**, the electrical system **40** also includes a booster module **52** electrically connected to the batteries **45** and to the rectifier and inverter modules **42**, **44**. The booster module **52** is configured to compensate for undervoltage of the power supplied by the batteries **45**. The booster module **52** comprises a booster bridge **53** which may include a plurality of chokes.

(38) The rectifier module **42**, the inverter module **44**, the internal static bypass switch **46** and the booster module **52** include different types of semiconductors. For instance, as shown in FIGS. **15** and **16**, the rectifier module **42**, the inverter module **44** and the booster module **52** include a plurality of semiconductors **115**. The semiconductors may be of any suitable type. In this embodiment, the semiconductors **115** of the rectifier module **42**, the inverter module **44** and the booster module **52** are insulated-gate bipolar transistors (IGBT). The semiconductors **115** of the rectifier module **42** and the inverter module **44** are comprised by rectifier-inverter bridges **48**. The semiconductors **115** of the booster module **52** are comprised by the booster bridge **53**. Furthermore, the internal static bypass switch **46** also has a plurality of semiconductors (not shown) which are thyristor modules. The semiconductors of the rectifier and inverter modules **42**, **44**, of the internal static bypass switch **46** and of the booster module **52** are commonly known and manufactured for example by Semikron©. Other different types of semiconductors are also contemplated.

(39) As can be seen, in FIGS. **19** and **20**, in this embodiment, each semiconductor **115** has a generally rectangular upper surface. Moreover, each semiconductor **115** has two connectors at opposite ends thereof which allow the semiconductor **115** to be secured to a liquid cooling device as will be described in more detail below. Notably, each connector is configured to receive a fastener therein.

(40) It is contemplated that the electrical system **40** could be configured differently in other embodiments. For instance, in other embodiments, the UPS **10** may be of a type other than a double conversion online UPS and therefore the electrical system **40** may include additional electrical components or some components may be omitted.

(41) Furthermore, the skilled person will understand that the electrical system **40** includes various other electrical components which are not described here for brevity, namely as the specific configuration of the electrical system **40** is not of particular importance in the context of the present

application and therefore this description of the electrical system **40** is not meant to be exhaustive. For instance, other electrical components such as a backfeed protection device, filters, chokes, capacitors, fuses, etc. are also part of the electrical system **40**.

(42) Various of the above-described components of the electrical system **40** generate a significant amount of heat during use. Consequently, in order to prevent overheating of the components (which may otherwise negatively affect their performance), the UPS **10** has a cooling system that combines both liquid cooling and air cooling. More specifically, the UPS **10** is cooled by (i) circulating ambient air through the interior space **15** of the UPS **10** to evacuate heated air therefrom via the two air-to-liquid heat exchangers **90**; and (ii) routing water to a plurality of liquid cooling devices **50** disposed in the interior space **15** of the UPS **10** to cool certain components of the electrical system **40**.

(43) The manner in which the air-to-liquid heat exchangers **90** ensure the air cooling of the UPS **10** will now be described. Both air-to-liquid heat exchangers **90** are identical and therefore only one of the air-to-liquid heat exchangers **90** will be described herein with reference to FIG. **14**. It is understood that the description applies to both air-to-liquid heat exchangers **90**.

(44) The air-to-liquid heat exchanger **90** is configured to cool air flowing therethrough. The air-to-liquid heat exchanger **90** includes a cooling coil **92** (shown in dashed lines in FIG. **14**) for circulating fluid therethrough. In particular, in this embodiment, water is circulated through the cooling coil **92**. To that end, the cooling coil **92** has an inlet **96** and an outlet **98** to respectively feed water to and discharge water from the cooling coil **92**. As will be explained in greater detail below, the cooling coil **92** is fluidly connected to a corresponding one of the pumping modules **60**. The air-to-liquid heat exchanger **90** also includes a plurality of fins **94** that are in thermal contact with the cooling coil **92**. The fins **94** are spaced apart from one another to allow air to flow therebetween and thus through the air-to-liquid heat exchanger **90**. The air-to-liquid heat exchanger **90** has a frame **95** supporting the cooling coil **92** and the fins **94**.

(45) In this embodiment, the air-to-liquid heat exchanger **90** is mounted to a corresponding one of the doors **30**. Notably, the frame **95** of the air-to-liquid heat exchanger **90** is fastened to the exterior side **36** of the door **30**. The air-to-liquid heat exchanger **90** is disposed in a position vertically higher than the corresponding pumping module **60**. As will be explained in more detail below, this vertically higher position of the air-to-liquid heat exchanger **90** facilitates the circulation of air through the UPS **10**. Moreover, the air-to-liquid heat exchanger **90** is aligned with a door opening **91** (shown in dashed lines in FIG. **3**) defined by the door **30** so that air can flow from the interior space **15** through the door opening **91** and through the air-to-liquid heat exchanger **90**.

(46) In order to force air through the air-to-liquid heat exchanger **90**, in this embodiment, as shown in FIG. **7**, a plurality of fans **102** are mounted to each door **30**, on the interior side **34** thereof. In particular, in this embodiment, twelve fans **102** are mounted to each door **30** and are aligned with the corresponding door opening **91** such that, when the fans **102** are activated, they force air through the door openings **91** and through the air-to-liquid heat exchangers **90**. As can be seen, when the doors **30** are closed, the fans **102** are contained within the housing **12**. In this embodiment, the fans **102** are relatively small in size and a rotation axis of the impeller of each fan **102** extends generally horizontally.

(47) As designated by arrows **97** in FIG. **4**, operation of the fans **102** causes air to be pulled into the interior space **15** of the housing **12** through the top wall **16** and the bottom wall **18** and forced out through the door openings **91** and the air-to-liquid heat exchangers **90**. In particular, the top wall **16** defines a plurality of top openings **23** (FIG. **5**) which, in this embodiment, are slits. Similarly, the bottom wall **18** defines a plurality of bottom openings (not shown) in the shape of slits. Ambient air is thus drawn into the interior space **15** through the top openings **23** and the bottom openings of the top wall **16** and the bottom wall **18**. The air drawn into the interior space **15** absorbs some of the heat generated by the UPS **10** and is then forced out through the door openings **91** and through the air-to-liquid heat exchangers **90**, where heat is transferred from the air to the water being circulated

through the cooling coil **92**. Thus, the air discharged on the other side of the air-to-liquid heat exchanger **90** is cooled and therefore a cool ambient temperature can be maintained in the surroundings of the UPS **10** (in this case the data center). As will be understood, the positioning of the air-to-liquid heat exchangers **90** vertically higher than the pumping modules **60** on the doors **30** facilitates pushing out air (through the air-to-liquid heat exchangers **90**) incoming from the top openings **23** and the bottom openings of the top wall **16** and the bottom wall **18** with an adequate distribution of the air flow as the air-to-liquid heat exchangers **90** are approximately at mid-height of the doors **30**.

(48) As mentioned above, and as shown in FIGS. **9** and **10**, the UPS **10** includes the plurality of liquid cooling devices **50** enclosed in the interior space **15** of the housing **12** for cooling different electrical components of the electrical system **40**. In particular, in this embodiment, the liquid cooling devices **50** are mounted to the semiconductors of the rectifier module **42**, the inverter module **44** and the internal static bypass switch **46** for cooling thereof. Notably, as the semiconductors generate heat during operation thereof, each liquid cooling device **50** absorbs heat from a corresponding semiconductor and evacuates the heat via fluid circulated through a fluid conduit **125** defined by the liquid cooling device **50** (described in greater detail further below). Such liquid cooling devices **50** are also commonly referred to as “cold plates” or “water blocks” and can use different types of fluids for transferring heat out of the liquid cooling device. In this embodiment, water is circulated through the liquid cooling devices **50** to evacuate heat therefrom. It is contemplated that other fluids may be used instead of water (e.g., dielectric fluid).

(49) It should be understood that the liquid cooling devices **50** may be mounted to electrical components other than the semiconductors. Notably, the liquid cooling devices **50** may be mounted to various other heat-generating electrical components of the electrical system **40**, including for example chokes.

(50) The liquid cooling devices **50** will be described in greater detail further below.

(51) The liquid cooling devices **50** define part of an internal fluid circuit C**1** through which water is circulated locally at the UPS **10**. Conversely, in this embodiment, in use, the cooling coils **92** of the air-to-liquid heat exchangers **90** define part of an external fluid circuit C**2** through which water is circulated at least in part externally of the UPS **10**, namely through external cooling equipment **75** (shown schematically in FIGS. **11** and **13**). The external cooling equipment **75** can be any type of equipment configured to cool the water being routed thereto from the cooling system of the UPS **10**. For instance, in this example of implementation, the external cooling equipment **75** is a dry cooler installed in an exterior of the data center (e.g., on the roof thereof).

(52) Since the internal fluid circuit C**1** is provided to feed water to the liquid cooling devices **50**, the internal fluid circuit C**1** may alternatively be referred to as a “liquid cooling circuit”. Similarly, since the external fluid circuit C**2** is provided to feed water to the air-to-liquid heat exchangers **90**, the external fluid circuit C**2** may alternatively be referred to as an “air cooling circuit”.

(53) In order to circulate water through the liquid cooling circuit C**1** (including the liquid cooling devices **50**), two pumping modules **60** are provided. Notably, as mentioned above, each pumping module **60** is mounted to a corresponding one of the two doors **30** and disposed on the exterior side **36** thereof. In this embodiment, both pumping modules **60** are configured identically, namely being a mirror image of one another. Thus, only one of the pumping modules **60** will be described in detail herein. It is to be understood that the description applies to both pumping modules **60**.

(54) With reference to FIGS. **11** and **13**, the pumping module **60** includes two pumps **62** that are fluidly connected to the liquid cooling devices **50** for circulating water through the liquid cooling circuit C**1** of the UPS **10**. It is contemplated that fewer or more pumps **62** may be provided in other embodiments (e.g., a single pump **62**). In this embodiment, the two pumps **62** are of a relatively small size to accommodate their positioning on the corresponding door **30** since the door **30** has a limited surface area for placing the pumping module **60**. Notably, the pumps **62** are fluidly connected in series and thus functionally could be replaced by a larger, more powerful pump.

However, due to restrictions imposed by the size of the door **30**, in this embodiment, two pumps **62** are provided instead of a larger one. Furthermore, two pumps **62** fluidly connected adds a level of redundancy to the pumping module **60**, notably as if one of the pumps **62** stops working, the second pump **62** can continue ensuring the flow of water in the liquid cooling circuit C**1**.
(55) The pumping module **60** also includes two plate heat exchangers **64**, fluidly connected in series, for transferring heat from water circulating in the liquid cooling circuit C**1** to water circulating in the air cooling circuit C**2**. Thus, in use, the plate heat exchangers **64** define part of both the liquid cooling and air cooling circuits C**1**, C**2** (without an actual exchange of water between both circuits C**1**, C**2** occurring at the plate heat exchangers **64**). As shown in FIG. **13**, in the liquid cooling circuit C**1**, the plate heat exchangers **64** are fluidly connected to the pumps **62** and to the liquid cooling devices **50**. In the air cooling circuit C**2**, the plate heat exchangers **64** of the pumping module **60** are fluidly connected to the cooling coil **92** of the corresponding air-to-liquid heat exchanger **90** (provided on the same door **30** as the pumping module **60**) and to the external cooling equipment **75**. It is contemplated that fewer or more plate heat exchangers **64** could be provided in other embodiments (e.g., a single plate heat exchanger).
(56) With reference to FIG. **12** which shows a simplified representation of one of the plate heat exchangers **64**, each plate heat exchanger **64** includes a plurality of plates **65** stacked with one another and defining gaps **67** therebetween for circulation of fluid between the plates **65**. Each plate heat exchanger **64** has two inlets and two outlets, with one inlet and one outlet corresponding to each circuit C**1**, C**2**. Each plate **65** defines apertures to fluidly connect a first sub-set of the gaps **67** separately from a second sub-set of the gaps **67**, with each sub-set of gaps **67** defining part of a corresponding one of the liquid cooling circuit C**1** and the air cooling circuit C**2**. The gaps **67** of each of sub-set of gaps **67** are arranged in alternating fashion such that a gap **67** defining part of one of the circuits C**1**, C**2** is adjacent to a gap **67** defining part of an other one of the circuits C**1**, C**2**. Heat is thus transferred from one of the circuits C**1**, C**2** to an other one of the circuits C**1**, C**2** via the plates **65**. As will be understood, water from the liquid cooling and air cooling circuits C**1**, C**2** does not mix in the plate heat exchanger **64**. Such plate heat exchangers **64** are known and therefore will not be described in further detail herein.
(57) As shown in FIGS. **11** and **13**, the pumping module **60** also includes an expansion tank **66** fluidly connected to the liquid cooling circuit C**1**. The expansion tank **66** compensates for pressure in the liquid cooling circuit C**1** as a function of temperature variations. The expansion tank **66** may be fluidly connected to the liquid cooling circuit C**1** at different locations. A pressure reducing valve **68** is fluidly connected between a junction J**1** of the liquid cooling circuit C**1** and a junction J**2** of the air cooling circuit C**2**. The junction J**1** is disposed between the liquid cooling devices **50** and the pumps **62** such that water heated at the liquid cooling devices **50** passes through the junction J**1** prior to arriving at the pumps **62**. The junction J**2** is disposed between the plate heat exchangers **64** and external cooling equipment **75** that defines part of the air cooling circuit C**2**. Water heated at the plate heat exchangers **64** in the air cooling circuit C**2** passes through the junction J**2** prior to arriving at the external cooling equipment **75**. The pressure reducing valve **68** ensures that makeup water is added to the liquid cooling circuit C**1** when the pressure in the liquid cooling circuit C**1** drops below a certain value (e.g., 1.5 bars). The pressure reducing valve **68** is omitted in embodiments in which the liquids used in the liquid cooling circuit C**1** and the air cooling circuit C**2** are different.
(58) A pressure relief valve **70** is also fluidly connected between a junction J**3** of the liquid cooling circuit C**1** and a junction J**4** of the air cooling circuit C**2** to open the liquid cooling circuit C**1** to the air cooling circuit C**2** when the pressure in the liquid cooling circuit C**1** exceeds a certain value (e.g., 3 bars). The junction J**3** is disposed between the pumps **62** and the plate heat exchangers **64** such that water pumped by the pumps **62** passes through the junction J**3** prior to reaching the plate heat exchangers **64**. The junction J**4** is disposed between the junction J**2** and the external cooling equipment **75**. The pressure relief valve **70** is omitted in embodiments in which the liquids used in

the liquid cooling circuit C**1** and the air cooling circuit C**2** are different.
(59) The pumping module **60** also includes a strainer **72** for filtering the water in the air cooling circuit C**2** before it enters the plate heat exchangers **64**. The strainer **72** also filters water in the liquid cooling circuit C**1** when the liquid cooling circuit C**1** is initially being filled up (during setup of the cooling system of the UPS **10**). In some embodiments, an additional strainer with higher filtration capacity may be installed between the junction J**2** and the pressure reducing valve **68** to enhance the filtration of water in the liquid cooling circuit C**1** when the liquid cooling circuit C**1** is initially being filled up.
(60) As shown in FIG. **11**, in this embodiment, the components **62**, **64**, **66**, **68**, **70**, **72** of the pumping module **60** are mounted to a base frame **74** which is connected to the exterior side of the corresponding door **30**. The base frame **74** includes a planar panel **76** and a plurality of tabs **78** extending generally perpendicular to the panel **76**. The various components **62**, **64**, **66**, **68**, **70**, **72** of the pumping module **60** are supported by the tabs **78**. In particular, the panel **76** and the tabs **78** of the base frame **74** are made from a single sheet of metallic material which is cut and bent into shape. The base frame **74** may be made in different ways in other embodiments. In addition, as shown in FIGS. **1** to **3**, a cover panel **80** is removably connected to the base frame **74** to cover the components of the pumping module **60** that are supported by the base frame **74**.
(61) Returning now to FIG. **13**, the liquid cooling circuit C**1** fluidly connects the pumps **62**, the plate heat exchangers **64** and the liquid cooling devices **50** to one another. In particular, in the liquid cooling circuit C**1**, water from the liquid cooling devices **50** flows to the pumps **62** (via piping **71**, FIGS. **1**, **2**) where the water is pumped to the plate heat exchangers **64** for cooling. The pumps **62** thus ensure the circulation of water through the liquid cooling circuit C**1**. At the plate heat exchangers **64**, the water routed thereto by the pumps **62** is cooled as heat is transferred from the water in the liquid cooling circuit C**1** to the water routed to the plate heat exchangers **64** in the air cooling circuit C**2**. Thus, in the liquid cooling circuit C**1**, cooled water is routed from the plate heat exchangers **64** to the liquid cooling devices **50** (via piping **73**, FIGS. **1** to **3**) so that the components of the electrical system **40** to which the liquid cooling devices **50** are mounted can be cooled. This process thus continually loops as heated water is routed from the liquid cooling devices **50** to the pumps **62** and so forth. Meanwhile, the air cooling circuit C**2** fluidly connects the plate heat exchangers **64** of the pumping module **60**, the external cooling equipment **75** and the corresponding air-to-liquid heat exchanger **90**. In particular, in the air cooling circuit C**2**, cooled water is discharged by the external cooling equipment **75** and flows from the external cooling equipment **75** to the inlet **96** of the cooling coil **92**. Water then flows through the cooling coil **92** where heat is transferred from the air flowing between the fins **94** (and thus through the air-to-liquid heat exchanger **90**) to the water in the cooling coil **92**. Then, water flows from the outlet **98** of the cooling coil **92** to an inlet of one of the plate heat exchangers **64**. In the plate heat exchangers **64**, heat is transferred from water in the liquid cooling circuit C**1** to water in the air cooling circuit C**2**. Thus, in the air cooling circuit C**2**, heated water is discharged via an outlet of the second consecutive plate heat exchanger **64** and routed to the external cooling equipment **75**. This process thus continually loops as the water cooled by the external cooling equipment **75** is routed back to the cooling coil **92** of the air-to-liquid heat exchanger **90**.
(62) The piping extending between the main piping of the UPS **10** (that is connected to the various components of the pumping modules **60** and to the air-to-liquid heat exchangers **90**) and the liquid cooling devices **50** is flexible piping having quick-connect fittings so as to facilitate installation and maintenance thereof, in addition to increasing safety by reducing the chance of a leak in the piping system.
(63) Other equipment may also define part of the circuits C**1**, C**2** but has not been described for simplicity. For instance, the air cooling circuit C**2** is also defined in part by one or more pumps (not shown) to ensure circulation of water in the air cooling circuit C**2**. These pumps are not present in the UPS **10** in this implementation, but rather form part of the data center installation as the air

cooling circuit C**2** can also be used in part to route water to other equipment in the data center (e.g., servers) for cooling thereof.

(64) While the liquid cooling circuit C**1** has been described in respect of a single one of the pumping modules **60** for simplicity, it is to be understood that, in this embodiment, the liquid cooling circuit C**1** of the UPS **10** is defined by the above-described components of both pumping modules **60**. Notably, as best shown in FIG. **10**, T-shaped or Y-shaped pipe fittings **77** are provided along the liquid cooling circuit C**1**, in fluid communication with the piping **71**, **73**, to distribute water from both pumping modules **60** to all of the liquid cooling devices **50** and to distribute water from all of the liquid cooling devices **50** to both pumping modules **60**. Furthermore, two pumping modules **60** fluidly connected adds a level of redundancy, notably since if one of the pumping module **60** has to stop working or to be removed for maintenance, the second pumping module **60** continues ensuring the flow of water in the liquid cooling circuit C**1**.

(65) Similarly, while the air cooling circuit C**2** has been described in respect of a single one of the pumping modules **60** for simplicity, it is to be understood that, in this embodiment, the air cooling circuit C**2** is defined by the above-described components of the both pumping modules **60** and by both air-to-liquid heat exchangers **90**. Notably, while each air-to-liquid heat exchanger **90** is fluidly connected to the plate heat exchangers **64** of the corresponding pumping module **60** (mounted to the same door **30**), water in the portion of the air cooling circuit C**2** defined by the plate heat exchangers **64** of both pumping modules **60** is merged in a same conduit to be routed to the external cooling equipment **75**. Moreover, both air-to-liquid heat exchangers **60** are fluidly connected to the external cooling equipment **75** to receive cool water therefrom.

(66) Liquid cooling of the electrical components of the UPS **10**, via the liquid cooling devices **50** and the pumping modules **60**, significantly dissipates the heat produced by the UPS **10**. For instance, between 70% to 90% of the heat generating capacity of the UPS **10** can be dissipated by the liquid cooling provided by the liquid cooling devices **50**. Consequently, the air cooling of the UPS **10** via the air-to-liquid heat exchangers **90** needs only dissipate a small proportion of the heat generating capacity of the UPS **10**. This allows reducing the size of the air cooling system of the UPS **10** compared to conventional UPSs, thus optimizing the usage of space within the data center. Notably, conventional UPSs are typically cooled by forced air convection alone and therefore the air cooling system must be powerful and large enough to handle all of the heat generating capacity of the UPS. For instance, in some cases, conventional UPSs rely on an air handler disposed between UPSs for cooling thereof or a large computer room air conditioning (CRAC) unit to cool the room within which the UPSs are located.

(67) Providing the pumping modules **60** and the air-to-liquid heat exchangers **90** on the doors **30** of the UPS **10** as described above can offer various advantages. Notably, the pumping modules **60** and its pumps **62** can be kept compact and locally accessible at the UPS **10** rather than having a large pumping system to circulate water to all of the UPSs of a data center. Moreover, the pumping modules **60** and the air-to-liquid heat exchangers **90** are easy to access which can facilitate their maintenance. For instance, the components of the pumping modules **60**, including for example the pumps **62**, as well as the air-to-liquid heat exchangers **90** can be removed from the UPS **10** from the exterior of the UPS **10**. Similarly, the position of the fans **102** on the doors **30** allows easy access thereto. In addition, positioning the pumping modules **60** outside of the interior space **15** of the UPS **10** allows ensuring that any potential leaks at the pumping modules **60** will not adversely affect the components of the electrical system **40**, thus ensuring the safety of the UPS **10** and the data center in general. Similarly, the position of the air-to-liquid heat exchangers **90** on the exterior sides of the doors **30** allows the connection between the external cooling equipment **75** and the UPS **10** to be outside of the UPS **10**, thereby preventing any potential leaks at the connection to affect function of the UPS **10**. This can facilitate the process of obtaining safety certification for the UPS **10** despite its implementation of a liquid cooling circuit that extends partly within the interior space **15**.

(68) Furthermore, as will be understood, placing the pumping modules **60** and the air-to-liquid heat exchangers **90** on the front side of the UPS **10**, in particular on the exterior sides **36** of the doors **30**, allows positioning two UPSs **10** side-by-side and thereby keep the surface area within the data center required to house and service the UPSs **10** relatively small. In addition, this manner of cooling the UPS **10** foregoes placing any equipment atop the UPS **10** (i.e., above the top wall **16** of the housing **12**) which can allow the UPS **10** to be installed in facility with low ceilings as only a small clearance above the top wall **16** has to be accommodated for air to be aspirated into the interior space **15** through the top wall **16**. In contrast, conventional UPSs often require placing an air handling unit laterally between two UPSs to circulate air therethrough, with the air handling unit occupying a significant amount of space. Moreover, conventional UPSs often require installing containment barriers above the UPSs to form “hot” and “cold” aisles to prevent hot air to be mixed with the cold air that is to be circulated through the UPSs. Besides using considerable space, this can present a challenge in terms of ceiling height as well as cable management for the UPSs. Furthermore, the hot and cold aisles formed by the containment barriers typically require implementing raised a floor and false ceiling to allow cooled air and hot air to circulate, which can be eliminated with the present technology
(69) In addition, the above-described configuration may facilitate retrofitting a conventional UPS to improve its cooling capacity and provide cooling autonomy thereto. Notably, a door assembly including the door **30** and the associated pumping module **60** and air-to-liquid heat exchanger **90** may be provided on its own to retrofit a conventional UPS. In particular, the target components of the conventional UPS that are intended to be cooled can be removed for mounting the liquid cooling devices **50** thereto (conventional heat sinks may have to be removed if applicable). Once the liquid cooling devices **50** are installed, the target components are reinstalled. The conventional door of the UPS is removed and replaced with the door assembly including the door **30** and the associated pumping module **60** and air-to-liquid heat exchanger **90** (the two doors may be replaced if desired). The piping is then connected to ensure the circuits C**1**, C**2** are established.
(70) The electrical power capacity of the UPS **10** may also be increased by the implementation of the cooling system as described above. Notably, the electrical power capacity of the UPS **10** may be increased by up to 10% by using the above-described cooling system, while using electrical components that are the same as those used in conventional UPSs. Indeed, a component that is watercooled has a lower operating temperature than the same component that is aircooled when its electrical capacity remains the same. Consequently it is also possible to increase the electrical capacity of a watercooled component while keeping its temperature below the temperature reached by this component when it is to be aircooled.
(71) The autonomy gained by the UPS **10** by placing the pumping modules **60** and air to liquid heat exchangers **90** on the UPS **10** also provides more flexibility in placing the UPS **10** within the data center. Notably, the UPS **10** can be placed in a same room as server racks containing servers and need not be isolated in a dedicated “power room” as is often the case with conventional UPSs.
(72) It should be noted that in some embodiments, only one pumping module **60** and one air-to-liquid heat exchanger **90** could be provided on one of the doors **30**. However, the inclusion of two pumping modules **60** and two air-to-liquid heat exchangers **90** provides redundancy to the cooling system of the UPS **10** as one of the pumping modules **60** and/or one of the air-to-liquid heat exchangers **90** could be disabled or removed for maintenance while the other pumping module **60** and the other air-to-liquid heat exchanger **90** still ensures some degree of cooling of the UPS **10**.
(73) In some embodiments, the two pumping modules **60** may define two separate and independent liquid cooling circuits C**1** (the T-shaped or Y-shaped pipe fittings **77** would be omitted), with a first liquid cooling circuit C**1** feeding water to some (e.g., half) of the fluid conduits **125** of each liquid cooling device **50**, and a second liquid cooling circuit C**1** feeding water to the remaining (e.g., the other half) of the fluid conduits **125** of the liquid cooling devices **50**. While this may not provide redundancy between the two pumping modules **60** as described above, if may provide redundancy

at the local level of each liquid cooling device **50** as the fluid conduits **125** thereof are fed by two fluidly independent liquid cooling circuits.

(74) The configuration and functioning of the liquid cooling devices **50** will now be described in greater detail with reference to FIGS. **17** to **22**. Each liquid cooling device **50** is configured to cool a plurality of target components **115**. In the illustrated example, the target components **115** are the semiconductors of any of the components of the electrical system **40** of the UPS **10**. Notably, in this example of implementation, each cooling device **50** is configured to cool three semiconductors **115**. The target components **115** may be any other suitable components that generate heat and could benefit from cooling. Since each liquid cooling device **50** is identical in this example, only one of the liquid cooling devices **50** will be described in detail herein. It is to be understood that the same description applies to the other liquid cooling devices **50**.

(75) As best shown in FIG. **20**, the liquid cooling device **50** includes a base member **110** and a plurality of cover members **112** connected to the base member **110**. As will be described in more detail below, the cover members **112** define, together with the base member **110**, a plurality of fluid conduits **125**. The fluid conduits **125** define part of the liquid cooling circuit C**1**. In this embodiment, the fluid conduits **125** of the liquid cooling device **50** are independent from one another in that water flows through the fluid conduits **125** in parallel. As such, each fluid conduit **125** receives, at an inlet thereof, cool water that has not circulated through another one of the fluid conduits **125** (as would be the case if they were connected in series). In other embodiments, some of the fluid conduits **125**, such as those that are configured to overlap a same one of the target components **115**, could be connected in series such that water flows through one of the fluid conduits **125** and successively through another one of the fluid conduits **125**.

(76) The base member **110** and the cover members **112** are made of copper so as to effectively conduct heat. Other thermally conductive materials are also contemplated.

(77) As shown in FIGS. **17** and **18**, the base member **110** of the liquid cooling device **50** is in thermal contact with the three semiconductors **115** that are intended to be cooled by the liquid cooling device **50**. More specifically, a lower surface **114** of the base member **110** (shown in FIG. **19**), on a lower side **116** of the base member **110**, is placed in contact with the semiconductors **115**. The lower surface **114** is generally flat to ensure proper contact between the semiconductors **115** and the lower surface **114**. A thermal paste may be disposed between the lower surface **114** and the semiconductors **115** to efficiently transmit heat from the semiconductors **115** to the base member **110**. The base member **110** defines a plurality of fastener openings **113** configured to receive respective fasteners (not shown) to fasten the semiconductors **115** to the base member **110**. The fastener openings **113** extend from the lower side **116** to an upper side **118** of the base member **110**.

(78) In this embodiment, the base member **110** is generally rectangular. The base member **110** is sized to span the three semiconductors **115**. A thickness of the base member **110**, which provides rigidity to the base member **110**, is dependent on a surface area of the base member **110**. In this embodiment, the thickness of the base member **110** is between 8 mm and 15 mm inclusively. More specifically, the thickness of the base member **110** is approximately 10 mm. It is desirable to provide the thinnest base member **110** while ensuring adequate rigidity thereof for the given surface area of the base member **110** since providing a thinner base member **110** reduces the costs of production of the liquid cooling device **50**.

(79) As shown in FIGS. **21** and **22**, on its upper side **118** (opposite the lower side **116**), the base member **110** defines a plurality of pockets **120** that are spaced apart from one another. The pockets **120** are generally rectangular, and in particular generally square, and receive therein respective ones of the cover members **112**. Each pocket **120** is defined by a pocket upper surface **124** and side walls **126** (shown in FIG. **21**).

(80) In this embodiment, the number of pockets **120** defined by the base member **110** corresponds to the number of cover members **112** of the liquid cooling device **50**. Notably, the base member **110** defines an even number of pockets **120**, namely six pockets **120**, to receive respective ones of the

six cover members **112**. It is contemplated that a different number of pockets **120** could be defined by the base member **110** in other embodiments (e.g., if fewer or more semiconductors are to be cooled by the liquid cooling device **50**). For instance, in some embodiments, if a single semiconductor **115** is to be cooled by the liquid cooling device **50**, the base member **110** may define only two pockets **120**.

(81) The pockets **120** are arranged so that, when the liquid cooling device **50** is mounted to the semiconductors **115**, each of the three semiconductors **115** is at least partly overlapped by (i.e., disposed vertically above or below) a corresponding pair of the pockets **120**. Each pair of pockets **120** overlapping a given semiconductor **115** will be referred to herein as a pocket pairing **121**. In this embodiment, since the semiconductors **115** are arranged generally parallel to one another and spaced apart from one another, the pockets **120** are arranged in a rectangular array.

(82) With reference to FIG. **22**, in order to form the fluid conduits **125**, the base member **110** defines a plurality of fluid path recesses **122** on the upper side **118**. The fluid path recesses **122** define the path of each fluid conduit **125** through which water in the liquid cooling circuit C**1** flows to absorb heat from the semiconductors **115**. Each of the pocket upper surfaces **124** defines one of the fluid path recesses **122** such that each fluid path recess **122** is disposed in a respective one of the pockets **120** and, as shown in FIG. **23**, is aligned with a corresponding semiconductor **115**. The base member **110** thus defines six fluid path recesses **122**. In this embodiment, each fluid path recess **122** is shaped identically and thus only one of the fluid path recesses **122** in one of the pockets **120** will be described herein, with reference also being made to the fluid path recess **122** disposed in the other pocket **120** of the same pocket pairing **121**.

(83) The fluid path recess **122** has a circular inlet zone **130** and a circular outlet zone **132** which correspond to the points at which the fluid path recess **122** (and the corresponding fluid conduit **125**) respectively receives and discharges water. At the inlet zone **130**, the fluid path recess **122** splits into two separate channels **134** which merge together again at the outlet zone **132**. Each of the channels **134** defines a sinusoidal pattern along a majority of a span thereof. That is, each one of the channels **134** has a repetitive pattern approximating that of a sinusoidal function along at least half of the span of that channel **134**.

(84) The inlet zone **130** of the fluid path recess **122** is located closer to the middle plane MP than the outlet zone **132** (i.e., a distance between the inlet zone **130** and the middle plane MP is smaller than a distance between the outlet zone **132** and the middle plane MP). In other words, water enters the fluid path recess **122** closer to the middle plane MP than it is discharged from the fluid path recess **122**. Furthermore, the inlet zone **130** is generally centered between the side walls **126** (of the corresponding pocket **120**) that extend perpendicularly to the middle plane MP. As such, a central axis CA extending centrally between the side walls **126** of the given pocket pairing **121** that extend perpendicularly to the middle plane MP extends through the inlet zones **130** of the fluid path recesses **122** in both pockets **120** of the given pocket pairing **121**. Conversely, the outlet zone **132** is offset from the central axis CA and is thus closer to one of the side walls **126** that extend perpendicularly to the middle plane MP. The outlet zones **132** of the fluid path recesses in the pockets **120** of the same pocket pairing **121** are located on opposite sides of the central axis CA.

(85) The positions of the inlet and outlet zones **130**, **132** have been expressly chosen to have an optimal cooling effect on the corresponding semiconductor **115**. Notably, because water flowing through the fluid path recess **122** is coldest as it enters at the inlet zones **130**, the inlet zones **130** are positioned to be aligned with the regions of the corresponding semiconductor **115** that are subject to the highest temperatures during operation of the UPS **10**. Moreover, as water flowing through the fluid path recess **122** is hottest as it is discharged at the outlet zones **132**, the outlet zones **132** of the fluid path recesses **122** aligned with one of the semiconductors **115** are positioned specifically on opposite sides of the central axis CA. A thermal analysis of the implementation of these configurations of the fluid path recesses **122** has demonstrated to offer an optimal heat dissipation for each semiconductor **115**.

(86) The fluid path recesses **122** may be shaped differently in other embodiments, such as in embodiments in which components other than semiconductors are the target components that the liquid cooling device **50** is intended to cool.
(87) In this embodiment, the channels **134** of the fluid path recesses **122** have a width of approximately 2 mm and a depth of approximately 4 mm. Notably, the fluid path recesses **122** are milled into the pocket upper surfaces **124**. This can simplify and accelerate the production of the liquid cooling devices **50**, notably as various base members **110** can be mounted to a computer numerically controlled (CNC) mill and the six fluid path recesses **122** of each base member **110** can be machined in a single operation.
(88) Returning to FIG. **20**, each cover member **112** is received in a corresponding one of the pockets **120** to define, together with the corresponding fluid path recess **122**, a respective fluid conduit **125**. The cover members **112** are planar plates which are shaped and sized to be congruous with the shape of the pockets **120**. Notably, in this embodiment, the cover members **112** are generally square with rounded corners. Each cover member **112** has a lower side (not shown) and an upper side **140**. A lower surface (not shown) of each cover member **112** is planar and faces the fluid path recess **122** of the corresponding pocket **120** to define the corresponding fluid conduit **125** together therewith. Each cover member **112** has a fluid inlet and a fluid outlet for receiving and discharging water through the corresponding fluid conduit **125** defined by that cover member **112**. The fluid inlet and fluid outlet of the cover member **112** are defined by an inlet opening and an outlet opening respectively. An inlet tube **142** and an outlet tube **144** are connected to each cover member **112**. In particular, the inlet tube **142** is welded to each cover member **112** and is fluidly connected to the corresponding inlet opening defined by the cover member **112**. Similarly, the outlet tube **144** is welded to each cover member **112** and is fluidly connected to the corresponding outlet opening defined by the cover member **112**. In this embodiment, the inlet tubes **142** and the outlet tubes **144** are welded to the cover members **112** via an autogenous welding process. Notably, the inlet tubes **142** and outlet tubes **144** are welded to the cover members **112** via laser welding (also referred to as “laser beam welding”). Namely, as will be described in more detail below, autogenous welding processes such as laser welding offer advantages in terms of quality which is particularly important to ensure safety in the context of feeding a liquid into the interior space **15** of the UPS **10** which has many electrical components that could be negatively affected by a leak.
(89) When the cover members **112** are in position in the corresponding pockets **120**, the fluid inlet and the fluid outlet of each cover member **112** are respectively aligned with the inlet zone **130** and the outlet zone **132** of the fluid path recess **122** disposed in the corresponding pocket **120**.
(90) As can be seen in FIG. **20**, the cover members **112** are relatively thin. Notably, a thickness of each cover member **112** is between 2 mm and 5 mm inclusively. In particular, in this embodiment, the thickness of each cover member **112** is approximately 3 mm. This thinness of the cover members **112** allows using a small amount of material for their manufacture which makes the liquid cooling device **50** more affordable to produce. This is particularly advantageous considering that many liquid cooling devices **50** are included in the UPS **10**.
(91) This thinness of the cover members **112** poses a challenge in terms of connecting the cover members **112** to the base member **110**. Notably, in order to reduce a number of parts of the liquid cooling device **50**, it is desirable to weld the cover members **112** to the base member **110** since fastening the cover members **112** to the base member **110** via fasteners (e.g., screws, bolts, rivets, etc.) would require adding sealing members (e.g., gaskets) to prevent leaks. However, because the cover members **112** and the base member **110** are thin, welding the cover could potentially warp the material of the cover members **112** which could result in leaks during use of the liquid cooling device **50**. To address this, the cover members **112** have a relatively small periphery such that each cover member **112** can be quickly welded along its periphery to the side walls **126** of the corresponding pocket **120**. Notably, at least partly for this reason, the liquid cooling device **50** is designed such that each semiconductor **115** is cooled by two fluid conduits **125** defined by two

small separate cover members **112** rather than a larger cover member spanning an equivalent surface area. This limits the amount of heat that is absorbed by the cover members **112** during welding thereof. Moreover, a pause time between welding each cover member **112** consecutively allows the base member **110** to cool which can prevent it from warping. Furthermore, in this embodiment, the cover members **112** are connected to the base member **110** by an autogenous welding process. That is, the cover members **112** are welded to the base member **110** without adding material to form the welds. This is contrast with non-autogenous welding processes in which a filler metal is added to join components. In particular, in this embodiment, the cover members **112** are connected to the base member **110** by laser welding (similarly to the outlet and inlet tubes **142**, **144** described above). This ensures that the liquid cooling device **50** is properly sealed as the welds connecting the cover members **112** to the base member **110** are made of the same material as the cover members **112** and the base member **110** rather than relying on the quality of a filler metal. Moreover, with laser welding, the welds are not as susceptible to the presence of contaminants and the quality of the welds does not depend on the manner in which the welding material flows as is the case with non-autogenous welding processes.
(92) As can be seen, the cover members **112** (and the corresponding pockets **120**) are relatively small in size. Notably, in this embodiment, each cover member **112** and each pocket **120** extends along a surface area of approximately 20 cm.sup.2. As such, the periphery of the cover members **112** is relatively small which allows welding the cover members **112** without exposing the cover members **112** or the base member **110** to excessive heat during assembly which could result in the liquid cooling device **50** being defective as described above.
(93) In an alternative embodiment, with reference to FIG. **24**, a liquid cooling device **250** could be provided instead of the liquid cooling device **50**. The liquid cooling device **250** includes a base member **210**, a plurality of cover members **212** and a plurality of intermediate members **235** which are connected to one another. The base member **210** is similar to the base member **110** described above and the cover members **212** are similar to the cover members **112** described above and therefore they will not be described in detail herein. Each intermediate member **235** is received in a corresponding pocket of the base member **210**. Each cover member **212** is stacked atop one of the intermediate members **235**. The addition of the intermediate members **235** allows defining, for each pocket defined by the base member **210**, two separate fluid conduits that are fluidly connected in parallel (not in series) to provide redundancy in case one of the fluid conduits should be disabled (e.g., blocked). Notably, for each pocket of the base member **210**, a lower fluid conduit is defined between the base member **210** and the corresponding intermediate member **235**, and an upper fluid conduit is defined between the intermediate member **235** and the corresponding cover member **212**. Each intermediate member **235** defines a fluid inlet and a fluid outlet to receive fluid into and discharge fluid from the lower fluid conduit. A liquid cooling device having upper and lower fluid conduits of this type is described in greater detail in European Patent Application No. 18315027.5, filed on Sep. 4, 2018, the entirety of which is incorporated herein by reference. In some embodiments in which the UPS **10** is equipped with the liquid cooling devices **250**, the two pumping modules **60** may define two separate and independent liquid cooling circuits C**1** (the T-shaped or Y-shaped pipe fittings **77** would be omitted), with a first liquid cooling circuit C**1** feeding water to some lower (or upper) fluid conduits of some liquid cooling devices **250**, and a second liquid cooling circuit C**1** feeding water to the corresponding upper (or lower) fluid conduits of these liquid cooling devices **250**.
(94) Returning now to FIGS. **17**, **18** and **20**, the liquid cooling device **50** also includes an inlet manifold **160** fluidly connected to the fluid inlets of the cover members **112** via the inlet tubes **142**, and an outlet manifold **162** fluidly connected to the fluid outlets of the cover members **112** via the outlet tubes **144**. The inlet and outlet manifolds **160**, **162** are disposed exteriorly of the base member **110**, in particular being disposed above the base member **110**. The inlet manifold **160** is configured for feeding water to the fluid conduits **125**. Notably, the inlet manifold **160** receives

cooled water from the plate heat exchangers **64** of the pumping modules **60**. The outlet manifold **162** is configured for discharging water from the fluid conduits **125**. In particular, heated water discharged through the fluid outlets of the cover members **112** flows through the outlet manifold **162** and is routed back towards the pumps **62** of the pumping modules **60** and cooled again at the plate heat exchangers **64**.

(95) In this embodiment, the inlet manifold **160** and outlet manifold **162** are fluidly connected to each fluid conduit **125** in a manner so as to establish a Tichelmann loop (sometimes written "Tickelman") through the liquid cooling device **50**. More specifically, a distance between the inlet manifold **160** and the outlet manifold **162** through each fluid conduit **125** is approximately the same. This balances the fluid flow rates though the fluid conduits **125** of the liquid cooling device **50** so that they are approximately the same, which avoids having to implement valves to achieve similar flow rates through the fluid conduits **125**.

(96) As noted by the detailed configurations described and depicted above by the embodiments of FIGS. **21-23**, there is provided a plurality of pocket pairings **121**, in which the pocket pairings **121** are spaced apart along a lateral direction of the liquid cooling device **50**. Each of the pocket pairings **121** comprises individual pockets **120** that are juxtaposedly arranged in close proximity to each other. In turn, each of the pockets **120** is configured to accommodate a fluid conduit **125** for circulating a cooling liquid therethrough, in which the fluid conduit **125** incorporates an inlet zone **130** for receiving the cooling liquid and an outlet zone **132** for discharging the circulated fluid.

(97) Given that the disclosed embodiments provide an arrangement in which the individual pockets **120** of each pocket pairing **121** are in close proximity and that each of the individual pockets **120** incorporates a corresponding fluid conduit **125** requiring separate inlet and outlet coupling elements, namely the inlet and outlet tubes **142**, **144**, it may be desirable to minimize the number of coupling elements that receive the cooling liquid (e.g., inlet tubes **142**) and discharge the cooling liquid (e.g., outlet tubes **144**), namely in embodiments in which the two fluid conduits **125** defined in adjacent ones of the pockets **120** are connected in series.

(98) To this end, FIGS. **25**A, **25**B respectively depict perspective and cross-sectional views of another embodiment of the liquid cooling device **50** in which the liquid cooling device **50** defines an internal interconnecting channel **404** between the fluid conduits **125** of a pocket pair **121**. In particular, in this alternative embodiment, the base member **110** of the liquid cooling device **50** defines two juxtaposed pockets **120**A, **120**B, whereby respective fluid path recesses **122** are defined by the base member **110** within the pockets **120**A, **120**B. As shown, the interconnecting channel **404** is configured as a "slanted-shaped" internal liquid conveyance element disposed between the juxtaposed pockets **120**A, **120**B of the pocket pairing **121**. As such, the juxtaposed pockets **120**A, **120**B and the fluid conduits **125** defined therein are fluidly connected in series by the interconnecting channel **404**.

(99) In particular, the interconnecting channel **404** is formed between respective adjacent sidewalls of the juxtaposed pockets **120**A, **120**B. One end of the slanted-shaped interconnecting channel **404** is coupled to the outlet (e.g., **132**) of fluid conduit **125**A contained by pocket **102**A and the other end of interconnecting channel **404** is coupled to the inlet (e.g., **130**) of fluid conduit **125**B contained by pocket **102**B. With this configuration, the interconnecting channel **404** internally distributes the fluid from conduit **125**A of pocket **120**A to the conduit **125**B of pocket **120**B.

(100) As best indicated by FIG. **25**B, in this embodiment, the slanted-shaped interconnecting channel **404** extends at an angle relative to the lower surface **114** of the base member **110** such that one end of the interconnecting channel **404** is disposed vertically higher than the other end of the interconnecting channel **404**. Notably, this configuration may facilitate forming the interconnecting channel **404**, namely machining the interconnecting channel **404** such as by milling or drilling. For instance, during manufacturing, the base member **110** may be set an angle, with the lower surface **114** extending partly vertically such as to allow the material removing tool (e.g., milling cutter or drill bit) to reach the sidewall of one of the juxtaposed pockets **120**A, **120**B.

(101) Alternatively, FIGS. **26**A, **26**B respectively depict perspective and cross-sectional views of another embodiment of the liquid cooling device **50** in which the liquid cooling device **50** defines an internal interconnecting channel **404** between the fluid conduits **125** of a pocket pair **121**. In this embodiment, the interconnecting channel **404** is configured as a "V-shaped" internal liquid conveyance element disposed between the juxtaposed pockets **120**A, **120**B of the pocket pairing **121** to fluidly connect the pockets **120**A, **120**B in series.
(102) As best indicated by FIG. **26**B, in this embodiment, the V-shaped interconnecting channel **404** is formed by milling or drilling operations each of the respective adjacent sidewalls of the juxtaposed pockets **120**A, **120**B. The milling/drilling of the adjacent sidewalls are performed at similar height levels and at similar downward angles relative to the pockets **120**A, **120**B orientation such that the resulting cuts are aligned and connect with each other at a vertex point to form the V-shaped interconnecting channel **404**.
(103) It will be appreciated that, in furtherance to the slant-shaped and V-shaped profiles described above, the interconnecting channel **404** may comprise any suitable dimensions, shapes, profiles, and materials that effectively facilitate the transfer of fluids from one fluid conduit to another within pocket pairings.
(104) The concept of facilitating the transfer of fluids between fluid conduits of juxtaposed pockets may be expanded to multiple sets of such pockets within a cooling base member. As such, FIG. **27** depicts a liquid cooling base member **110** of another embodiment of the liquid cooling device **50** incorporating multiple liquid cooling elements with the interconnecting channels, in accordance with embodiments of the present disclosure.
(105) In the nonlimiting embodiment, liquid cooling base member **110** comprises an arrangement of multiple pockets, an overhead fluid inlet tube **142**, and an overhead fluid outlet tube **144**. The arrangement of pockets may comprise, for example, twelve pockets **120**A-**120**I in which each pocket accommodates fluid conduits **125**A-**125**I (shown schematically in FIG. **27**). The pockets that are closely positioned (or juxtaposed) may be grouped into three sets: (i) **120**A-**120**D; (ii) **120**E-**120**H; and (iii) **120**I-**120**L. The closely-positioned pockets within each set are fluidly coupled together by interconnecting channels **404** to circulate the cooling fluid throughout the respective fluid conduits **115** in series. As will be appreciated, in this embodiment, since the fluid conduits **125**A-**125**I are interconnected internally by respective interconnecting channels **404**, a single inlet tube **142** and a single outlet tube **144** are provided for the entirety of the fluid conduits **125**A-**125**I.
(106) As shown, certain pockets between the pocket sets, such as, for example, **120**A, **120**E, and **120**I may be designated as inlet pockets. The inlet pockets **120**A, **120**E, and **120**I also incorporate the interconnecting channels **404** to direct the fluid received by the overhead fluid inlet tube **142** to the inlet pockets **120**A, **120**E, and **120**I of the different pocket sets.
(107) Relatedly, certain pockets between the pocket sets, such as, for example, **120**D, **120**H, and **120**L may be designated as outlet pockets. The outlet pockets **120**D, **120**H, and **120**L also incorporate the interconnecting channels **404** to direct the discharge of the fluid from the outlet pockets **120**D, **120**H, and **120**L of the different pocket sets to the overhead fluid outlet tube **144**.
(108) FIG. **28** depicts a variation of the liquid cooling base member of FIG. **27** in accordance with another embodiment of the present disclosure. Notably, in this embodiment, the single inlet tube **142** and the single outlet tube **144** are connected to the base member **110** on a lateral side thereof (rather than to the respective cover members **112** that cover the corresponding pockets). As such, the liquid cooling device **50** can be made more compact, namely having a smaller height profile than if the overhead inlet and outlet tubes **142**, **144** were connected to the cover members **112**.
(109) The above-described configuration of the liquid cooling devices **50** is simple and cost-efficient to manufacture while ensuring an optimized cooling of the semiconductors **115**. Notably, the various separate fluid conduits **125** in each liquid cooling device **50** ensures that heat is dissipated from a majority and even an entirety of an upper surface of each semiconductor **115.** In particular, the base member **110** is in contact with the entire upper surface of each semiconductor

115 including a central portion thereof. This may not be the case for instance if each semiconductor **115** were instead cooled by two smaller separate liquid cooling devices (each defining a corresponding fluid conduit). Namely, the central portion of the semiconductor **115** would not be in contact with either of the liquid cooling devices in such a case, thereby decreasing the efficiency of heat dissipation of the semiconductor **15**. Moreover, the size of the base member **110** as to span various semiconductors **115** facilitates the positioning of the liquid cooling devices **50** in the UPS **10**. In addition, as discussed above, manufacturing of the liquid cooling devices **50** by machining the base members **110** and welding of the cover members **112** is greatly simplified while ensuring that the liquid cooling device **50** is safe to use in an electrically powered environment. It is contemplated that the configurations of the liquid cooling device **50** described herein can be used for heat-generating components other than a semiconductor and in other contexts than in a UPS.
(110) Modifications and improvements to the above-described embodiments of the present technology may become apparent to those skilled in the art. The foregoing description is intended to be exemplary rather than limiting. The scope of the present technology is therefore intended to be limited solely by the scope of the appended claims.

Claims

1. A liquid cooling device for cooling at least one target component, the liquid cooling device comprising: a base member configured to be in thermal contact with the at least one target component to be cooled by the liquid cooling device, the base member including a lower surface configured to be placed in contact with the at least one target component, the base member defining, on an upper side thereof, a plurality of pockets spaced apart from one another; the base member defining in part a plurality of fluid conduits configured to internally circulate a cooling liquid therethrough, each of the plurality of pockets arranged to accommodate a corresponding fluid conduit of the plurality of fluid conduits; an interconnecting channel structure defined by the base member and extending between two neighboring pockets of the plurality of pockets and configured to distribute the cooling liquid from a fluid conduit of one pocket to a fluid conduit of a neighboring pocket; and a plurality of covers, wherein each cover is connected to the base member and is received in a pocket of the plurality of pockets, wherein a first cover of the plurality of covers comprises a fluid inlet for receiving the cooling liquid, and wherein a second cover of the plurality of covers comprises a fluid outlet for discharging the cooling liquid, wherein, the interconnecting channel structure comprises a first interconnecting channel and a second interconnecting channel such that, each of the plurality of pockets incorporates the first interconnecting channel to service neighboring pockets disposed along a first dimension of the base member and incorporates the second interconnecting channel defined by the base member to service neighboring pockets disposed along a second dimension of the base member.
2. The liquid cooling device of claim 1, wherein each of the fluid conduits comprises a fluid inlet to receive the cooling liquid and a fluid outlet to discharge the cooling liquid.
3. The liquid cooling device of claim 2, wherein the fluid outlet of a fluid conduit of a first neighboring pocket distributes the cooling liquid to the fluid inlet of a fluid conduit of a second neighboring pocket via the interconnecting channel structure.
4. The liquid cooling device of claim 1, wherein the interconnecting channel structure is formed between adjacent sidewalls of respective neighboring pockets.
5. The liquid cooling device of claim 1, wherein the interconnecting channel structure comprises a slanted-shaped profile in which one end of the first interconnecting channel is disposed vertically higher than another end of the first interconnecting channel.
6. The liquid cooling device of claim 1, wherein the interconnecting channel structure comprises a V-shaped profile in which both ends of the first interconnecting channel are disposed vertically higher than a downwardly-angled vertex point.

7. The liquid cooling device of claim 1, wherein the plurality of pockets comprises an even number of pockets arranged along a two dimensional plane of the base member.
8. The liquid cooling device of claim 1, wherein: the first cover comprises an overhead fluid inlet tube for supplying the cooling liquid to the plurality of pockets from a liquid source and the second cover comprises an overhead fluid outlet tube for expelling the cooling liquid back to the liquid source.
9. The liquid cooling device of claim 1, wherein the base member comprises a lateral fluid inlet tube, disposed along a side surface of the base member, for supplying the cooling liquid to the plurality of pockets from a liquid source and a lateral fluid outlet tube, disposed along a side surface of the base member, for expelling the cooling liquid back to the liquid source.
10. The liquid cooling device of claim 1, wherein the interconnecting channel structure is formed by a milling or drilling process.
11. The liquid cooling device of claim 1, wherein the plurality of pockets are distributed in a grid formation, the grid formation having a plurality of rows and a plurality of columns.
12. A liquid cooling device for cooling at least one target component, the liquid cooling device comprising: a base member configured to be in thermal contact with the at least one target component to be cooled by the liquid cooling device, the base member including a lower surface configured to be placed in contact with the at least one target component, the base member defining, on an upper side thereof, a plurality of pockets spaced apart from one another; the base member defining in part a plurality of fluid conduits configured to internally circulate a cooling liquid therethrough, each of the plurality of pockets arranged to accommodate a corresponding fluid conduit of the plurality of fluid conduits, wherein each fluid conduit of the plurality of fluid conduits is shaped in a generally sinusoidal pattern within the plurality of pockets; and the base member defining in part an interconnecting channel structure extending between two neighboring pockets of the plurality of pockets and configured to distribute the cooling liquid from a fluid conduit of one pocket to a fluid conduit of a neighboring pocket, wherein, the interconnecting channel structure comprises a first interconnecting channel and a second interconnecting channel such that, each of the plurality of pockets incorporates the first interconnecting channel to service neighboring pockets disposed along a first dimension of the base member and incorporates the second interconnecting channel defined by the base member to service neighboring pockets disposed along a second dimension of the base member.
13. The liquid cooling device of claim 12, wherein a fluid outlet of a fluid conduit of a first neighboring pocket distributes the cooling liquid to a fluid inlet of a fluid conduit of a second neighboring pocket via the interconnecting channel structure.